



DEPARTMENT OF ENERGY
NAVAL REACTORS REPRESENTATIVE'S OFFICE
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01 May 2026

From: R. A. Crossley, Radiological and Environmental Controls Lead, Puget Sound
To: Benjamin Beeler, Ph.D.

Subj: COMPLETION OF NE 533 EXAMINATION #4 ICO IVAN DAVIS

Ref: (a) Exams for EOL e-mail correspondence dtd 06 Feb 2026

1. This memorandum documents completion of Nuclear Fuels Performance (NE 533) Examination #4 conducted for Ivan Davis on the 1st of May 2026. The examination was provided to me on the 23rd of April 2026 via e-mail from Professor Benjamin Beeler. The examination was proctored by me as prescribed in reference (a).

a. The examination was conducted at the address listed at the top of this letterhead.

b. The examination was conducted in a location that afforded the examination candidate an adequate opportunity to conduct the examination free of distraction.

c. The examination was conducted with the supplied equation sheet and blank paper. The examination candidate used a model TI-30XS calculator during the examination. No other materials were afforded to the candidate.

d. The examinee was afforded 75 minutes to conduct the examination which occurred between the hours of 1300 to 1415. A standard wall clock in the room was used to indicate time allotted to the examination candidate.

2. Following completion of the examination, I retained custody of the examination and all associated paperwork used during the examination as required by reference (a).

3. At no time did the examination candidate breach the academic integrity standards.

4. Any questions or concerns can be submitted to me by phone at 360-476-2829 or via email at raaid.a.crossley.mil@us.navy.mil.

A handwritten signature in blue ink, appearing to read "R. A. Crossley", is located at the bottom right of the page.

R. A. CROSSLEY

Exam 4: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

Equations for NE533

$$\frac{\delta^n}{2} = K^n t \rightarrow \delta = K t^{1/n}$$

$n \approx 4$ for Zr alloys

$$\delta^*(\mu m) = 5.1 \exp\left(\frac{-550}{T}\right)$$

$$t^*(d) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{T}\right)$$

$$K_L\left(\frac{\mu m}{d}\right) = 7.48 \times 10^6 \exp\left(\frac{-12500}{T}\right)$$

$$\delta(\mu m) = \delta^* + K_L(t - t^*)$$

$$w = (m_2 - m_1)/S \quad (mg/dm^2)$$

$$\delta(\mu m) = \frac{w}{14.7}$$

$$C_H^{clad}(wt.ppm) = \frac{2f \times \delta \times \rho_{oxide} \times f_{ZrO_2}^O \times \frac{M_H}{M_O}}{(t - \frac{\delta}{PBR}) \times \rho_{metal}} \times 10^6$$

PG 1

① GIVEN:

IN RX FOR 500 DAYS @ 625K

$$W_{Ti} = 600 \mu\text{m}$$

$$f_{H_2} = 14\% \rightarrow 0.14$$

$$P_{BZ} = 1.56$$

$$P_{Zr} = 6.5 \text{ g/cc}$$

$$P_{Zr} = 5.68 \text{ g/cc}$$

$$a.) \delta^* = 5.1 e^{-550/T} = 5.1 e^{(-550/625)} = 2.115 \mu\text{m}$$

$$t^* = 6.62 \times 10^{-7} e^{11949/K} = 6.62 \times 10^{-7} e^{(11949/625)} = 133.01 \text{ d}$$

500 > 133.01 d $\Rightarrow$ POST-TRANSITION REGIME

$$K_L = 7.48 \times 10^6 e^{-12500/T} = 7.48 \times 10^6 e^{(-12500/625)} = 0.0154 \mu\text{m/d}$$

$$\delta = \delta^* + K_L(t - t^*) = 2.115 \mu\text{m} + 0.0154 \mu\text{m/d}(500 - 133.01 \text{ d})$$

$$\boxed{\delta = 7.77 \mu\text{m}}$$

$$b.) C_{H_2}^{\text{CALD}} \text{ WT. PPM} = \frac{2f \times \delta \times P_{O_2} \times f_{ZrO_2} \times \frac{M_H}{M_O}}{(t - \frac{\delta}{P_{BZ}}) \times P_{\text{METAL}}} \times 10^6$$

$$f_{ZrO_2} = \frac{32}{91+32} = 0.2602$$

$$\frac{M_H}{M_O} = \frac{1}{16}$$

$$C_{H_2}^{\text{CALD}} \text{ WT. PPM} = \frac{2(0.14) \times (7.77) \times (5.68) \times (0.2602) \times (\frac{1}{16})}{(600 - \frac{7.77}{1.56}) \times (6.5)} \times 10^6$$

$$\boxed{C_{H_2}^{\text{CALD}} = 51.96 \text{ WT. PPM}}$$

$$W = 14.7\% = 7.77 \cdot 14.7 = 114.2$$

$$114.2 \text{ mg/dm}^2 = 1.142 \times 10^{-3} \text{ g/cm}^2$$

$$m_H = 0.14 \left(\frac{2(1.142 \times 10^{-3})}{16} \right)$$

$$m_H = 1.999 \times 10^{-5} \text{ g/cm}^2 ?$$

PG 2

- ② FOR AQUEOUS CORROSION OF Zr CLADDING, THE LIMITING STEP FOR RATE OF CORROSION IS THE TRANSPORT OF OXYGEN SPECIES THROUGH THE OXIDE LAYER.
- ③ HYDRIDES FORM IN CLADDING @ COOLER AREAS WITH TENSILE STRESS BECAUSE H_2 MIGRATES DOWN A TEMP. GRADIENT VIA THE SORET EFFECT. IN ORDER FOR A HYDRIDE TO START, ZrO_2 REACTIONS MUST OCCUR TO ENTER CLADDING AT A HIGH SOLUBILITY RATE.
- ④ A RIA IS A REACTIVITY INITIATED ACCIDENT IN WHICH POSITIVE REACTIVITY OCCURS CAUSING $P_{RX} \uparrow$ QUICKLY ALONG WITH T_f OF THE FUEL. IN A PWR, THIS WOULD BE A CONTROL ROD EJECTION OR UNCONTROLLED OUTWARD ROD MOTION. IN A BWR THE RIA IS FROM A CONTROL ROD DROP EVENT RESULTING IN CTRL BLADES BEING DROPPED OUT OF THE CORE.

DURING A RIA, $P_{RX} + TEMP \uparrow$ SIGNIFICANTLY RESULTING IN FUEL EXPANSION RAPIDLY, FISSION GAS RELEASE RESULTING IN FUEL ROD INTERNAL PRESSURE $\uparrow$ ALL OF WHICH INCREASES STRAIN ON THE CLADDING. IF CHF IS EXCEEDED, DNB OR DRYOUT OCCURS $\Rightarrow$ CLADDING TEMP RAPIDLY $\uparrow$ $\Rightarrow$ CLADDING MELT OR BURST OCCURS.

- ⑤ A LOCA IS A LOSS OF COOLANT ACCIDENT IN WHICH COOLANT FLOW IS LOST, USUALLY FROM A MECHANICAL SYSTEM FAILURE CREATING A COOLANT LEAK. DURING A LOCA, THE LOSS OF COOLANT PREVENTS REMOVING DECAY HEAT FROM THE FUEL RODS OR CELLS $\Rightarrow$ FUEL ROD TEMPS BEGIN TO RISE. THIS RESULTS IN INCREASED CLADDING OXIDATION + HYDRIDING AND HIGH CLADDING TEMPS WHICH LEAD TO CLADDING FAILURE.

IN A RIA THE ACCIDENT IS RAPID, WHEREAS A LOCA IS A SLOW INCREASE IN FUEL ROD/CELL TEMPERATURES.

5/1/2026

Pg 3

- ⑥ BURUP WILL ↑ ACCIDENT VULNERABILITY SINCE REDUCED FUEL + CLADDING WILL FAIL @ LOWER POINTS COMPARED TO NEW FUEL ROD ASSEMBLIES. AS BURUP OCCURS, FISSION PRODUCT GXS WILL ↑ INSIDE FUEL ROD, FUEL CONDUCTIVITY ↓ AND CORROSION AND THE CLADDING WILL ↑. AS OXIDE LAYERS ↑, H_2 PICKUP ↑ ⇒ HYDRIDING ↑ + CLADDING DUCTILITY ↓. THEREFORE BURUP WILL ↑ THE PROBABILITY OF AN ACCIDENT.
- ⑦ 4 PATHWAYS TO MAKE FUEL/CLADDING ACCIDENT TOLERANT:
 - a. IMPROVE CLADDING THROUGH ↑ STRESS RESISTANCE + MELTING POINTS.
 - b. IMPROVE THE FUEL BY ↑ MELTING MARGIN, ↓ FUEL DISPERSION, LOWER REQUIRED OPERATING TEMPS
 - c. IMPROVE FISSION PRODUCT RETENTION BY RETAINING GASES, SOLID OR NO FISSION PRODUCTS MORE EFFECTIVELY
 - d. IMPROVE STEAM RETENTION EFFECT BY ↓ OXIDATION RATES, ↓ H_2 GENERATION + HYDRIDE EMBELEMENT POTENTIAL.
- ⑧ A NEUTRON TERM AT = CONCEPT IS CRYSTALINE Zr Alloy CLADDING BECAUSE IT IS EASILY SWAPPED W/ EXISTING FUEL RODS/CERUS, AND ↓ OXIDATION AT HIGH TEMPERATURES SIGNIFICANTLY. LESS OXIDATION → LESS H_2 → LESS HYDRIDE EMBELEMENT.
- ⑨ 1D A HIGH TEMP STEAM EQUILIBRIUM, Zr CLADDING WILL REACT AS $Zr + 2H_2O \rightarrow ZrO_2 + 2H_2$. THIS REACTING INCREASES SIGNIFICANTLY ABOVE 1200°C IN WHICH THE Zr WILL TRANSITION FROM AN α -Zr TO A β -Zr WHICH WILL CAUSE A BRITTLE FRACTURE BEHAVIOR ON THE CLADDING.
- ⑩ THREE LIMITING PHENOMENA IN LWRs ARE:
 - a. CRYSTALLINE OXIDATION + H_2 PICKUP - OXIDE ↑ ⇒ H_2 ↑ ⇒ EMBELEMENT
 - b. DEGRADATION FROM NEUTRON SOILING IF CHF IS EXCEEDED, STEAM WILL BLUNNET THE HEAT TRANSFER SURFACE,
 - c. FUEL-CLADDING INTERFACING - FUEL EXPANSION INTO CLADDING ↑ HOOP STRESS WHICH ↑ SCC EFFECTS.

PG 4

AND SAFETY

- (11) CRUD EFFECTS FUEL PERFORMANCE[↑] BY LOWERING HEAT TRANSFER FROM FUEL ROD TO COOLANT WHICH $\uparrow$ FUEL TEMPS. IT CAN ALSO IMPACT COOLANT FLOW $\Rightarrow$ EFFECTING HEAT TRANSFER CHARACTERISTICS.

CRUD EFFECTS SAFETY IN THAT IT CAN DEPOSIT OUTSIDE THE CORE $\uparrow$ RADIATION LEVEL TO PERSONNEL.

- (12) TWO WATER CONTROLS IN LWRs ARE:

- PH CTRL USING BORIC ACID + LiOH - BORIC ACID AS A REACTIVITY CTRL MECHANISM + LiOH AS A PH MAINTAINER. ~~BY~~ BY CTRLING PH PROPERLY, CRUD IS $\downarrow$ IN THE CORE.
- IN BWRs NOBLE METAL INJECTION - THE NOBLE ~~METALS~~ METALS ARE USED TO IMPROVE H_2 CHEMISTRY TO $\downarrow$ ELECTROCHEMICAL CORROSION AND MINIMIZE INTERGRANULAR SCC.

- (13) TWO MICROSTRUCTURE PHENOMENA IN MOX:

- FUEL STRUCTURE & VOID FORMATION - AT HIGH TEMPS, THE FUEL PORES MOVE TO THE FUEL CENTER WHICH CREATES NEW GRAIN STRUCTURES WHICH MOVE OUTWARD TO THE MOST OUTER REGION.
- REDISTRIBUTION - PLUTONIUM & OXYGEN MOVE RADIIALLY DURING RX OPERATION. THIS CAN CAUSE AN ENRICHMENT EFFECT OF PU NEAR THE FUEL CENTER.
- SOG - FORMED AS Cs_2MoO_4 , IT IS A LAYER WHICH FORMS BETWEEN THE FUEL + CLADDING & $\downarrow$ FUEL SWELLING @ HIGH TEMPS.